CHILD AND ADOLESCENT PSYCHIATRY CLINIC · PROTOCOL

Stimulants and Growth

How we monitor and protect growth in children and adolescents taking ADHD medication.

Dr. Mohammed AlOtaibi (R4) · Child and Adolescent Psychiatry

1 BEFORE STARTING MEDICATION

01 Measure
Height on a stadiometer and weight

02 Plot
Both on a growth chart for her sex

03 History
Earlier heights from school or primary care

04 Parents
Both parents' heights for her expected height

05 Puberty
Stage, and whether periods have started

2 EXPECTED ADULT HEIGHT

Mid-parental height · Tanner

Girls

(Father + Mother − 13) ÷ 2

Target range: result **± 8.5 cm**

Example: father 172, mother 158 gives 158.5 cm, range 150 to 167 cm.

Boys

(Father + Mother + 13) ÷ 2

Target range: result **± 8.5 cm**

The ± 8.5 cm range covers about the 3rd to 97th percentile. Measure parents where possible, as reported heights are often inaccurate.

Calculator

FATHER, CM: 172

MOTHER, CM: 158

CHILD: Girl

158.5 cm

Target range 150.0 to 167.0 cm

3 MONITORING SCHEDULE

NICE NG87 · 1.8.5

Measure	Age 10 and under	Over 10 years	Who reviews
Height	Every 6 months	Every 6 months	Prescriber, on the growth chart, at every visit
Weight	Every 3 months	At 3 and 6 months, then every 6 months	

4 IF GROWTH CHANGES

Weight or appetite falling

- Give the dose with or after food
- Early breakfast and a late evening meal
- Dietitian for high-calorie, nutritious foods
- Consider a planned break or a change of medication

NICE 1.8.6

Height falling below expected

- Check height speed against parents' heights and puberty
- Rule out other causes: thyroid, coeliac disease, Turner syndrome
- Review the dose and consider a break over school holidays
- Refer to paediatric endocrinology

NICE 1.8.7

5 ADOLESCENT GIRLS

Before her first period
More growth is ahead. Watch height speed closely.

After her first period
About 6.6 cm usually remains. Most growth is done.

Eating
Screen for restrictive eating when appetite falls.

Misuse
Ask whether the medication is used to lose weight.

6 WHO DOES WHAT

NURSE	Height and weight at every visit, plotted
PSYCHIATRIST	Lowest effective dose, timing with meals, breaks or switches
DIETITIAN	Meal plan and eating screen
ENDOCRINE	Falling height or slow growth for her stage
FAMILY	Breakfast before the dose and a good evening meal

7 WHAT WE TELL THE FAMILY

"Most children on these medicines reach within a centimetre or two of the height they would have reached anyway. We will measure every six months and act early if growth slows."

Adapt to local policy. Referral follows clinical judgement and the growth chart.

Based on NICE NG87, 2018

EVIDENCE CARD · OCTOBER 2026

What the Evidence Says

Four clinical questions, each with a verdict and the studies behind it.

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QUESTION 1

Does it slow growth?

YES, EARLY AND MODESTLY

QUESTION 2

Does it reduce adult height?

USUALLY LITTLE OR NONE

QUESTION 3

Who is most at risk?

YEARS OF DAILY USE FROM EARLY CHILDHOOD

QUESTION 4

Can we reverse it?

WEIGHT YES, HEIGHT NO

1 DOES IT SLOW GROWTH?

Verdict · Yes, early and modestly

META-ANALYSIS

Height and weight percentiles fall on methylphenidate. Weight falls mostly in the first 12 months, height in the first 24 to 30 months. Dose, formulation and age did not change the effect.

Carucci 2021 · 18 studies · 4,868 children

META-ANALYSIS

The newest pooled analysis confirms small reductions in height, weight and BMI percentiles.

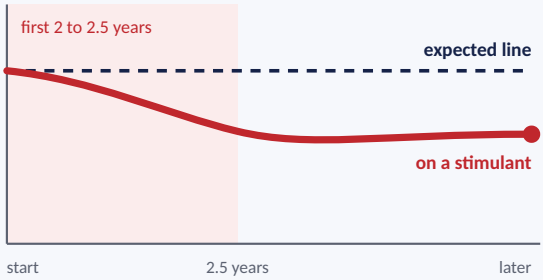
Cao 2026 · 33 studies

COHORT

ADHD itself is linked to temporary height deficits in early adolescence, unrelated to medication, that normalise later.

Spencer 1996 · 124 ADHD vs 109 controls

THE USUAL PATTERN ON TREATMENT



Schematic of the pooled pattern, not real data.

2 DOES IT REDUCE ADULT HEIGHT?

Verdict · Usually little or none

CONTROLLED COHORT

No difference in height growth over two years between treated and untreated ADHD. Puberty was not delayed.

ADDUCE 2023 and 2024 · 1,410 children, 27 European centres

BIRTH COHORT

No difference in adult height or in the size of the growth spurt.

Harstad 2014 · 340 ADHD vs 680 controls

NATIONAL COHORT

Adult height 167.8 vs 167.9 cm, children with and without ADHD.

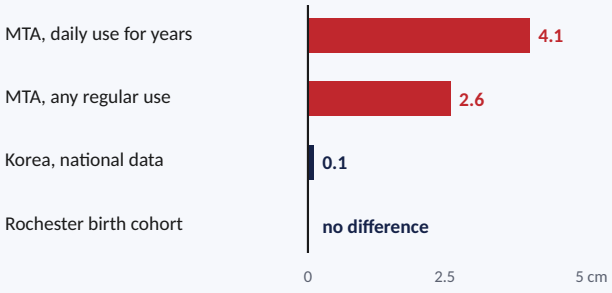
Song 2026, Korea · 34,850 young people

LONG-TERM TRIAL

Regular users were 2.6 cm shorter as adults than those with negligible use.

MTA, Swanson 2017 · 515 children followed to age 25

CENTIMETRES SHORTER AS ADULTS



Korea compares children with and without ADHD, not treated with untreated.

3 WHO IS MOST AT RISK?

Verdict · Long, early, high exposure

LONG-TERM TRIAL

The 53 children who took stimulants consistently for 16 years were 4.1 cm shorter as adults than negligible users.

MTA, Greenhill 2020

CHART REVIEW

Children started younger and taken to high doses lost more height percentile than those on usual doses.

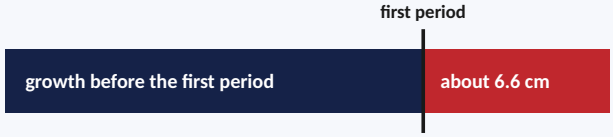
Ross 2020 · 52 high dose vs 118 usual dose

GROWTH DATA

Girls usually gain about 6.6 cm after their first period. Before it, more growth is at stake.

Post-menarcheal cohort, PubMed 38295778

WHY HER STAGE OF PUBERTY MATTERS



Before: more growth left, watch height speed.
After: most growth is done.

Average gain after the first period 6.6 ± 2.5 cm. Bar lengths illustrative.

4 CAN WE REVERSE IT?

Verdict · Weight yes, height no

RANDOMISED TRIAL

Children whose BMI fell on stimulants were given drug holidays, a daily 150 kcal supplement, or monthly monitoring. All improved weight gain. None improved height growth.

Waxmonsky 2020 · 230 children aged 5 to 12

OPEN-LABEL

On atomoxetine, a height shortfall peaked at 18 months and returned to expected by 24 months.

Spencer 2007 · 1,312 children

LONG-TERM TRIAL

Adults treated consistently in the MTA weighed more than their classmates. Watch weight in both directions.

MTA, Greenhill 2020

WHAT THE FIXES CHANGE

Strategy	Weight	Height
Drug holidays	Better	Same
Daily 150 kcal supplement	Better	Same
Monthly monitoring	Better	Same

Prevent weight loss early. Height does not catch up on request.

5 BOTTOM LINE FOR THE CLINIC

Expect

A small early slowing in the first two to three years.

Reassure

Most children reach close to their expected adult height.

Act early

Protect weight from the start, and measure height every six months.

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Most data are observational, and girls are under-represented.

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